

Augmentation

The purpose of **augmentation** is to provide nonparametric covariate adjustment for a marginal treatment contrast. An augmented estimator always has variance less than or equal to the original unadjusted estimator.

For treatment arms $k \in \{0, 1\}$, let Y_{ik} denote the outcome of subject i in arm k and X_{ik} a vector of covariates. Suppose $g(\theta_1, \theta_0)$ is a treatment contrast, and that the estimator of each θ_k admits an influence function expansion:

$$\hat{\theta}_k - \theta_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \psi(Y_{ik}; \theta_k) + o_p(1) = \frac{1}{n_k} \sum_{i=1}^{n_k} \psi_{ik} + o_p(1)$$

The unadjusted treatment contrast is:

$$\hat{\Delta}_{\text{unadj}} = g(\hat{\theta}_1, \hat{\theta}_0). \quad (1.1)$$

The adjusted treatment contrast takes the form:

$$\hat{\Delta}_{\text{adj}} = g(\hat{\theta}_1, \hat{\theta}_0) - \omega'(\bar{X}_1 - \bar{X}_0),$$

where ω is a weight vector:

$$\omega = \Sigma_{XX}^{-1} \Sigma_{X\Delta}.$$

Σ_{XX} is the variance of $\bar{X}_1 - \bar{X}_0$:

$$\Sigma_{XX} = \mathbb{V}(\bar{X}_1 - \bar{X}_0) = \mathbb{V}(\bar{X}_1) + \mathbb{V}(\bar{X}_0).$$

Letting $\hat{\mu}_X$ denote the grand mean:

$$\hat{\mu}_X = \frac{1}{n_1 + n_0} \left\{ \sum_{i=1}^{n_1} X_{i1} + \sum_{i=1}^{n_0} X_{i0} \right\},$$

Σ_{XX} may be estimated as:

$$\hat{\Sigma}_{XX} = \frac{1}{n_1^2} \sum_{i=1}^{n_1} (X_{i1} - \hat{\mu}_X)^{\otimes 2} + \frac{1}{n_0^2} \sum_{i=1}^{n_0} (X_{i0} - \hat{\mu}_X)^{\otimes 2}. \quad (1.2)$$

$\Sigma_{X\Delta}$ is the covariance between $\bar{X}_1 - \bar{X}_0$ and $g(\hat{\theta}_1, \hat{\theta}_0)$. To obtain this, consider the Taylor expansion of $g(\hat{\theta}_1, \hat{\theta}_0)$:

$$g(\hat{\theta}_1, \hat{\theta}_0) = g(\theta_1, \theta_0) + \dot{g}_1(\theta_1, \theta_0)(\hat{\theta}_1 - \theta_1) + \dot{g}_0(\theta_1, \theta_0)(\hat{\theta}_0 - \theta_0) + o_p(1).$$

Now:

$$\begin{aligned}\Sigma_{X\Delta} &= \mathbb{C}\{\bar{X}_1 - \bar{X}_0, g(\hat{\theta}_1, \hat{\theta}_0)\} = \mathbb{C}\{\bar{X}_1, g(\hat{\theta}_1, \hat{\theta}_0)\} - \mathbb{C}\{\bar{X}_0, g(\hat{\theta}_1, \hat{\theta}_0)\} \\ &= \dot{g}_1(\theta_1, \theta_0)\mathbb{C}(\bar{X}_1, \hat{\theta}_1 - \theta_1) - \dot{g}_0(\theta_1, \theta_0)\mathbb{C}(\bar{X}_0, \hat{\theta}_0 - \theta_0)\end{aligned}$$

Note that $\bar{X}_k$ only covaries with the effect estimate $\hat{\theta}_k$ from the same treatment arm.

Using the influence function expansions:

$$\begin{aligned}\Sigma_{X\Delta} &= \dot{g}_1(\theta_1, \theta_0) \cdot \frac{1}{n_1} \sum_{i=1}^{n_1} \mathbb{C}(\bar{X}_1, \psi_{i1}) - \dot{g}_0(\theta_1, \theta_0) \cdot \frac{1}{n_0} \sum_{i=1}^{n_0} \mathbb{C}(\bar{X}_0, \psi_{i0}) \\ &= \dot{g}_1(\theta_1, \theta_0) \cdot \frac{1}{n_1^2} \sum_{i=1}^{n_1} \mathbb{C}(X_{i1}, \psi_{i1}) - \dot{g}_0(\theta_1, \theta_0) \cdot \frac{1}{n_0^2} \sum_{i=1}^{n_0} \mathbb{C}(X_{i0}, \psi_{i0}).\end{aligned}$$

$\Sigma_{X\Delta}$ may be estimated as:

$$\hat{\Sigma}_{X\Delta} = \dot{g}_1(\hat{\theta}_1, \hat{\theta}_0) \cdot \frac{1}{n_1^2} \sum_{i=1}^{n_1} (X_{i1} - \hat{\mu}_X) \hat{\psi}_{i1} - \dot{g}_0(\hat{\theta}_1, \hat{\theta}_0) \cdot \frac{1}{n_0^2} \sum_{i=1}^{n_0} (X_{i0} - \hat{\mu}_X) \hat{\psi}_{i0}. \quad (1.3)$$

Overall, the covariate-adjusted estimator is:

$$\hat{\Delta}_{\text{adj}} = g(\hat{\theta}_1, \hat{\theta}_0) - \hat{\Sigma}'_{X\Delta} \hat{\Sigma}_{XX}^{-1} (\bar{X}_1 - \bar{X}_0). \quad (1.4)$$

The asymptotic variance is given by:

$$\mathbb{V}(\hat{\Delta}_{\text{adj}}) = \Sigma_{\Delta\Delta} - \Sigma'_{X\Delta} \Sigma_{XX}^{-1} \Sigma_{X\Delta}. \quad (1.5)$$

Here $\Sigma_{\Delta\Delta}$ is the variance of $g(\hat{\theta}_1, \hat{\theta}_0)$:

$$\Sigma_{\Delta\Delta} = \dot{g}_1^2(\theta_1, \theta_0) \mathbb{V}(\hat{\theta}_1) + \dot{g}_0^2(\theta_1, \theta_0) \mathbb{V}(\hat{\theta}_0) + o_p(1),$$

which may be estimated by:

$$\hat{\Sigma}_{\Delta\Delta} = \dot{g}_1^2(\hat{\theta}_1, \hat{\theta}_0) \cdot \frac{1}{n_1^2} \sum_{i=1}^{n_1} \hat{\psi}_{i1}^2 + \dot{g}_0^2(\hat{\theta}_1, \hat{\theta}_0) \cdot \frac{1}{n_0^2} \sum_{i=1}^{n_0} \hat{\psi}_{i0}^2. \quad (1.6)$$

Summary

The general procedure for nonparametric covariate adjustment via augmentation is:

1. Identify the treatment contrast of interest (1.1).
2. Calculate the influence function contributions of each subject (ψ_{ik}).
3. Calculate the variance components Σ_{XX} (1.2), $\Sigma_{X\Delta}$ (1.3), $\Sigma_{\Delta\Delta}$ (1.6).
4. Calculate the covariate-adjusted estimator (1.4) and its variance (1.5).

References

1. Jian F, Tian L, Fu H, Hasegawa T, Wei LJ. Robust Alternatives to ANCOVA for Estimating the Treatment Effect via a Randomized Comparative Study. *Journal of the American Statistical Association*. 2019; 114:528, 1854-1864.
doi.org/10.1080/01621459.2018.1527226.